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Near-Laurentian paleogeography of the Lawrence Head volcanics of central Newfoundland, northern Appalachians

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Abstract

Paleomagnetic analyses were completed on two volcanic units of the Exploits Group in Newfoundland's Central Mobile Belt, which are part of an Ordovician arc–back-arc system. The Tea Arm Volcanics display scattered and unstable characteristic directions that cannot be interpreted. However, stable end-points in six sites supported by great-circle analysis of seven sites in the mid-Arenigian to Llanvirnian Lawrence Head Volcanics yield a tilt- and strike-corrected characteristic direction of $D = 56°$, $I = 23°$, ($\alpha_{95} = 19°$, $k = 14$, $N = 6$). The magnetization is carried by magnetite and passes a tilt test. The corresponding paleolatitude of $12° \pm 10°$ is interpreted as southerly, and is similar to the paleolatitude of $11° \pm 4°$ for arc volcanics of the nearby Moreton's Harbour Group. However, this near-Laurentian paleolatitude is distinctly different from the paleolatitude of $31° \pm 8°$ reported for the Robert's Arm–(Cottrell's Cove)–Chanceport–Summerford volcanic terrane. The low paleolatitude of the Lawrence Head Volcanics supports the location of a major Ordovician subduction system near the Laurentian margin of Iapetus, whereas the present-day juxtaposition of latitudinally distinct elements in the area represents a complex accretionary history involving Early Silurian or older thrusting and younger strike-slip faulting of several Iapetan island-arc terranes to Laurentia.

1. Introduction

The interpretation of the Newfoundland Appalachians in the framework of the Wilson Cycle is well documented and long-standing (Dewey, 1969; Williams, 1979; Williams and Hatcher, 1983; van der Pluijm et al., 1993). In this scenario, western Newfoundland, which consists of Grenville basement overlain by shallow-water carbonates, represents the margin of Laurentia (Bradley, 1989) onto which ophiolites, such as the Bay of Islands complex, have been obducted (Stevens, 1970). Eastern Newfoundland is the type locality of the Avalon microcontinent, whose collision with Laurentia is responsible for the Siluro-Devonian Acadian orogenic pulse throughout the Canadian and New England Appalachians. Sandwiched between these two crustal blocks is the Central Mobile Belt (CMB) (Williams, 1964; van der Pluijm and van Staal, 1988) preserving Ordovician and older rocks that were formed within the Iapetus Ocean of Harland and Gayer (1972) (Fig. 1).

Several lines of evidence indicate that this ocean was relatively wide. Paleontological studies (McKerrow and Cocks, 1977; Neuman, 1984) provide evidence that it was extensive enough in the Early

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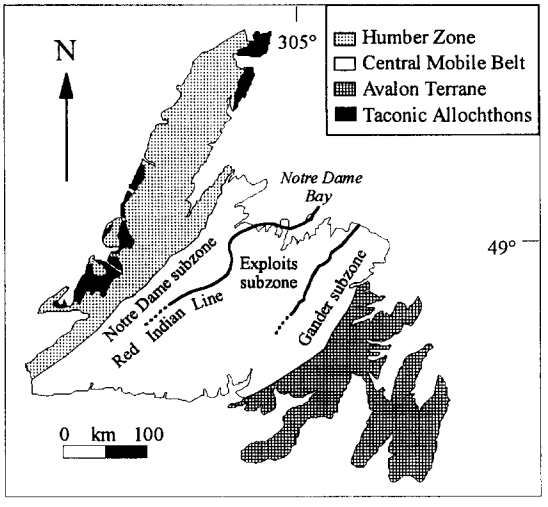


Fig. 1. Regional map of Newfoundland showing the general subdivision of two continental blocks and the intervening Central Mobile Belt (CMB), including the location of subzone boundaries within the northern portion of the CMB (after Williams et al., 1988).

Ordovician to prevent communication between fauna of Laurentian and Gondwanan affinity across the ocean. Furthermore, studies involving on- (Quinlan et al., 1992) and offshore (Keen et al., 1986) seismic reflection transects across Newfoundland show the presence of deep-crustal blocks corresponding to these two continents.

Within the CMB finer subdivisions are proposed to account for distinct regions within this wide ocean basin. Williams (1979) defined a Dunnage and a Gander zone, which represent Iapetus remnants and the Avalonian continental rise, respectively. Subsequently, Williams et al. (1988) and O'Brien (1991) proposed to subdivide the Dunnage zone into the Notre Dame and Exploits subzones, separated by the Red Indian Line, based on different geological and geophysical characteristics. The position of the latter boundary, however, is either in dispute (van der Pluijm et al., 1990; Van der Voo et al., 1991), or justification for its existence is questioned altogether (Dec and Swinden, 1994).

Paleomagnetism has shown to be a critical tool in determining the latitudinal relationships of presently-juxtaposed terranes in the Notre Dame Bay area of the northern Appalachians. Johnson et al. (1991) determined that the island arc that is now the Moreton's Harbour Group formed adjacent to the southern margin of Laurentia at $11°S \pm 4°$ south latitude. In van der Pluijm et al. (1990), it was proposed that the closure of the back-arc basin that separated this arc from Laurentia was responsible for the mid-Ordovician emplacement of ophiolites in western Newfoundland. Van der Voo et al. (1991) determined that the adjacent Robert's Arm, Chanceport and Summerford groups (RCS Terrane) formed at $31°S \pm 8°$, which implies that a major Ordovician boundary is located at today's site of the Lobster Cove–Chanceport Fault, though not necessarily coinciding with the currently exposed fault.

This study attempts to further refine the paleogeography of CMB tectonic elements prior to their accretion to Laurentia. It presents data from the Lawrence Head Volcanics of the Exploits Group, which is within several kilometers of the Moreton's Harbour sampling localities of Johnson et al. (1991), and the Cottrell's Cove Group (a Robert's Arm Group equivalent; Dean, 1978; Arnott et al., 1985; Bostock, 1988; Lafrance and Williams, 1992).

2. Regional geology

The Exploits Group is comprised of submarine mafic volcanic units at the base and top of the sequence with one (Helwig, 1969; Dean, 1978) or two (Helwig, 1967; O'Brien, 1990) intervening sedimentary formations. Following the usage of Helwig (1967) and O'Brien (1990), these units are, from south to north, the Tea Arm Volcanics, the Saunder's Cove Formation, the New Bay Formation, and the Lawrence Head Volcanics, all of which are unconformably overlain by the Strong Island Chert (Fig. 2). Conodonts from the top of the New Bay Formation (Hibbard et al., 1977) and graptolites from the base of the Strong Island Chert (Williams et al., 1992) bracket the age of the Lawrence Head Volcanics between mid-Arenigian and Llanvirnian (i.e., early Middle Ordovician). Correlation between the Tea Arm Volcanics and volcanic formations at the base of the Wild Bight Group suggests that the lowermost unit of the Exploits Group is no older than Early Ordovician.

Geochemical studies show a progression upward through this sequence from island-arc tholeiite to

back-arc basin signatures, leading Dec et al. (1992) to interpret the group as an island arc that later underwent rifting. The Tea Arm Volcanics display island-arc signatures, and alkaline basalts within the Strong Island Chert display back-arc basin signatures, implying that the Lawrence Head Volcanics formed during the later rifting event (B.H. O'Brien, pers. commun., 1993). Further evidence for this setting comes from the New Bay Formation, which lies conformably between the Tea Arm and Lawrence Head volcanics and is comprised of turbidites and thick olistostromes that, according to Hughes and O'Brien (1994), accumulated during back-arc rifting. Chert deposition at the top of the sequence is concurrent with a decrease and final cessation of volcanism. An arc–back-arc setting for the area is also supported by Swinden et al. (1990), who infer a similar scenario for the correlative Wild Bight Group to the west.

Sedimentary units overlying the Exploits Group (Point Leamington and Lawrence Harbour Formations) are separated from the Cottrell's Cove Group by the Luke's Arm Fault (Helwig, 1967; Dean and Strong, 1977). This fault forms part of the Red Indian Line as proposed by Williams et al. (1988), a major terrane boundary that juxtaposes formations of contrasting stratigraphy, lithology, structure, fauna, plutonism, lead isotope signatures, and magnetic and Bouguer anomalies. The Luke's Arm Fault is subvertical and bedding-parallel with a complex deformation history (Dean and Strong, 1977) that includes thrusting followed by folding and strike-slip motion. It merges with the Lobster Cove–Chanceport Fault [site of the Red Indian Line as repositioned in van der Pluijm et al. (1990) and Van der Voo et al. (1991)] to the east on New World Island. Work by Blewett and Pickering (1988) on the adjacent Point Leamington Formation suggested that strike slip motion along the Luke's Arm Fault occurred in the Acadian (Siluro-Devonian) and was dominantly sinistral, followed by later dextral shearing throughout the northern Appalachians in the late Devonian

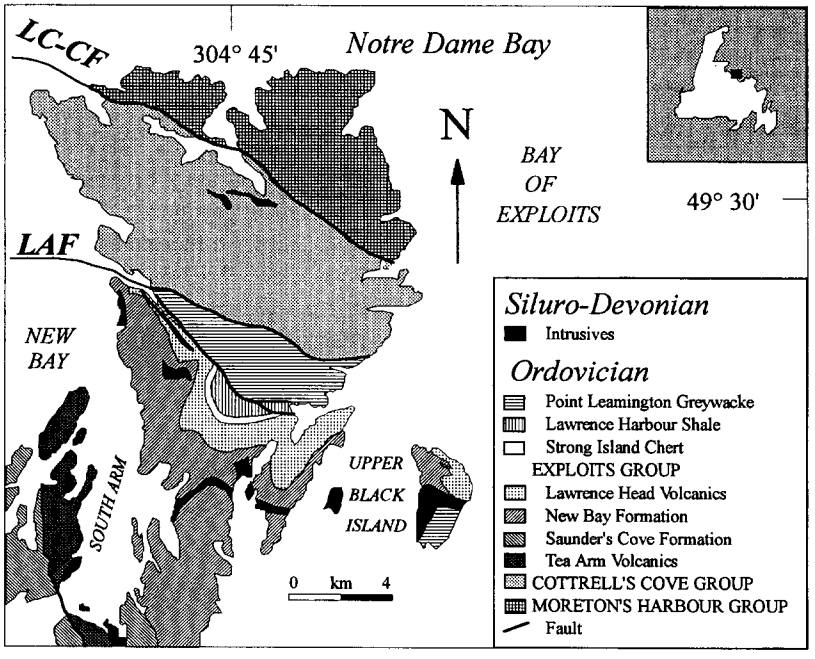


Fig. 2. Simplified geological map of the Fortune Harbour Peninsula of Notre Dame Bay (after O'Brien, 1990) showing the Moreton's Harbour and Exploits groups separated by the Cottrell's Cove Group, which is a correlative of the Robert's Arm, Chanceport, and Summerford Groups. The Cottrell's Cove Group is bounded to the south by the Luke's Arm Fault (*LAF*; part of the Red Indian Line as defined by Williams et al., 1988), and to the north by the Lobster Cove–Chanceport Fault (*LC–CF*; coinciding with the Red Indian Line as positioned by van der Pluijm et al., 1990, and Van der Voo et al., 1991).

to early Carboniferous. Whereas this corresponds well with much of the deformation throughout the northern Appalachians (see Hibbard, 1994), it seems in opposition to other studies elsewhere on the Luke's Arm Fault (Lafrance, 1989; Lafrance and Williams, 1992) that conclude dextral transpression active in the mid- to late quareSilurian. Regardless of the fault kinematics, the Luke's Arm Fault is a post-Ordovician strike-slip fault in the Newfoundland Appalachians that is unrelated to the paleogeography discussed in this paper. The timing of regional folding and cleavage development, which is critical for the interpretation of the paleomagnetic data, preceded or is mid- to late Silurian (Nelson, 1981; Pickering, 1987; Lafrance and Williams, 1992).

3. Sampling techniques

The Lawrence Head Volcanics were sampled at ten sites, Tea Arm Volcanics samples were collected at eighteen sites. Despite the greater number of sites in the Tea Arm Volcanics, a formation mean paleomagnetic direction cannot be presented because of the high degree of scatter in both declination and inclination within as well as between sites. Details of the study of the Tea Arm Volcanics are in Todaro (1994).

Lawrence Head volcanics sampling sites (Fig. 3) consist of shoreline exposures of basalt pillows except at site 5 (OLH5), where sedimentary strata were sampled. Pillows typically offer multiple exposed faces so that paleohorizontal could be determined from rounded pillow tops and tucked bottoms. Younging was determined in sediments, and by observing pillows shapes and their contact with sedimentary beds. Attitude measurements taken from sedimentary bedding and volcanic layering are in excellent agreement.

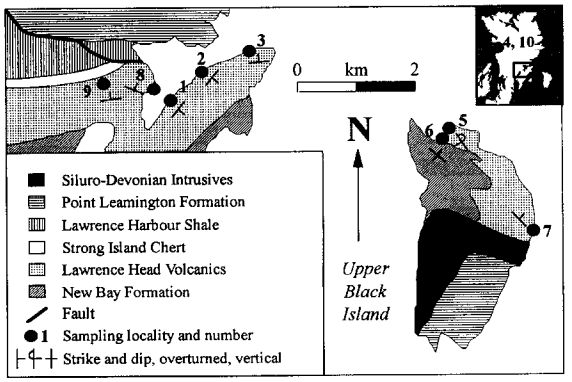


Fig. 3. Generalized geological map of the Lawrence Harbour area (modified from O'Brien, 1990) showing sampling sites in the Lawrence Head Volcanics. Note the two sites (OLH4 and OLH10) on the western shore of the Fortune Harbour Peninsula (inset).

Between ca. five and thirty standard 2.5-cm diameter paleomagnetic cores were collected at each site with a portable gas-powered drill. Typically two cores were taken from different faces of each pillow. These were spread from rims to centers and cover 5 to 15 m of section. Cores were oriented in-situ with a magnetic compass and inclinometer. The weak NRM intensities (1 to 200 mA/m) and the absence of noticeable compass deflections in the field indicate that local magnetic interference with attitude measurements was not significant.

4. Laboratory techniques

Standard samples (2.2 × 2.5 cm) were prepared from the oriented cores before storage and treatment in the magnetically shielded room at the University of Michigan. Low NRM intensities required that measurements be performed on either a ScT or 2G Enterprises cryogenic magnetometer. Pilot samples were stepwise thermally demagnetized with either a Schonstedt TSD-1 or Analytical Services TD-48 thermal demagnetizer, and companion samples were alternating field (AF) demagnetized with a Sapphire Instruments SI-4 single axis AF demagnetizer. Both methods often revealed similar demagnetization behavior, but thermal treatment was preferentially performed on the remaining samples. Data were analyzed with orthogonal vector endpoint diagrams (Zijderveld, 1967), and least-squares fit computer programs (Kirschvink, 1980; Torsvik, 1992) were used to determine the paleomagnetic vector components.

AF-treated samples from each site were used to determine the magnetic carriers and how they differ between sites. Stepwise isothermal remanent magnetization (IRM) acquisition up to 1.4 Tesla was performed with a water-cooled Varian electromagnet at

room temperature (25°C) to determine whole-sample behavior. Further information was gleaned from stepwise thermal treatment of magnetizations acquired in direct fields of 1.4 T, 0.5 T and 0.1 T along three mutually perpendicular axes. Subsequent matching of coercivity with unblocking temperature for grains magnetized along any axis allowed for inferences about the ferromagnetic minerals likely present and their relative proportions (Lowrie, 1990).

5. Paleomagnetic results

In the Lawrence Head Volcanics, 144 samples were collected from 10 sites throughout the formation where paleohorizontal and younging direction could be determined. Nearly all samples were taken from basalt pillows, with five samples taken from greywacke strata in a flysch succession located between the igneous rocks of site OLH5. The Lawrence Head Volcanics collection was pared down to seven sites (Tables 1 and 2) for reasons specified in the section on rejected sites. The variation in bedding orientation between sites allows for a tilt test; intrusives at site OLH7 were used for a baked contact test, but unstable magnetization of both intrusives and baked country rock precluded its usefulness.

5.1. Rejected sites

Three sampled sites were not used in the calculation of the formation mean direction. Samples from sites OLH5 displayed either weak NRM intensities (< 1 mA/m) with noisy and scattered demagnetization trajectories, or higher NRM intensities (between 2 and 10 mA/m) with univectorial demagnetization of the local present-day Earth's field direction. Samples from site OLH6 showed unstable demagnetization, whereas samples from site OLH8 displayed an unacceptably large scatter ($\alpha_{95} = 54.8$).

5.2. Demagnetization behavior

The NRM intensity of most samples varies between 1 and 200 mA/m. NRM directions vary both within and between sites (Fig. 4) and in almost all cases the resultant NRM is comprised of two components. The first corresponds to the local direction of the present-day geomagnetic field in in-situ coordinates, and is removed over differing thermal spectra depending upon the individual sample. This is likely a viscous remanence overprint. The second component begins to be removed at temperatures between 250°C and 500°C, and typically is completely removed by 585°C, indicating that low-titanium mag-

Table 1
Characteristic directions from the Ordovician Lawrence Head Volcanics, Central Newfoundland

Site	Bedding S/D	n/N	In-situ D/I	Tilt- corrected D/I	Strike- corrected D/I	k	α_{95}
OLH1	045/80	12/12	032/10	053/14	053/14	66.9	5.3
OLH2	045/88	11/11	030/15	060/15	060/15	17.6	11.2
OLH3	253/80	9/10	087/09	066/15	038/15	11.5	15.9
OLH4	320/80	9/23	360/37	08/−24	273/−24	38.0	8.5
OLH9	263/84	11/13	115/18	069/32	031/32	12.2	13.6
OLH10	317/99	9/15	160/−20	155/33	063/33	22.9	11.0
Mean			098/−01			1.7	78.7
Mean				088/33		2.7	50.5
Mean					056/23	14.1	18.5
Pole					031/55	$d_p = 10$	$d_m = 20$

Notes: S and D are the strike and dip of bedding (down dip to the right of strike). n and N are the number of samples used in the analysis and the number of samples collected at the site. D and I are the declination and inclination of the site mean direction, in degrees. k is the Fisher (1953) precision parameter. α_{95} is the 95% confidence cone around the mean direction.

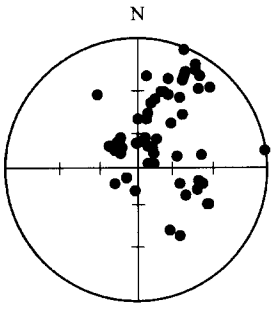


Fig. 4. Lower-hemisphere equal-angle projections displaying NRM directions of the Lawrence Head Volcanics.

netite is the magnetic carrier in these rocks. These two components combine to produce the (untreated) NRM directions displayed in Fig. 4.

Fig. 5 shows representative orthogonal vector endpoint diagrams in tilt-corrected coordinates for representative samples used in the mean direction calculation. Characteristic directions are determined from best-fit lines only; where unblocking temperatures for two components partly overlap, great-circle analysis was also used. Demagnetization diagrams for the first two sites (Fig. 5a and b) show simple behavior. The first component, which corresponds to the present-day field in in-situ coordinates, represents only a small portion of the total NRM and is removed below 490°C. The second component is defined by at least eight measurement steps and completely unblocks at or near 585°C. The sharp transition between the two vectors indicates that there is essentially no overlap in the thermal unblocking spectra of the grains carrying these two

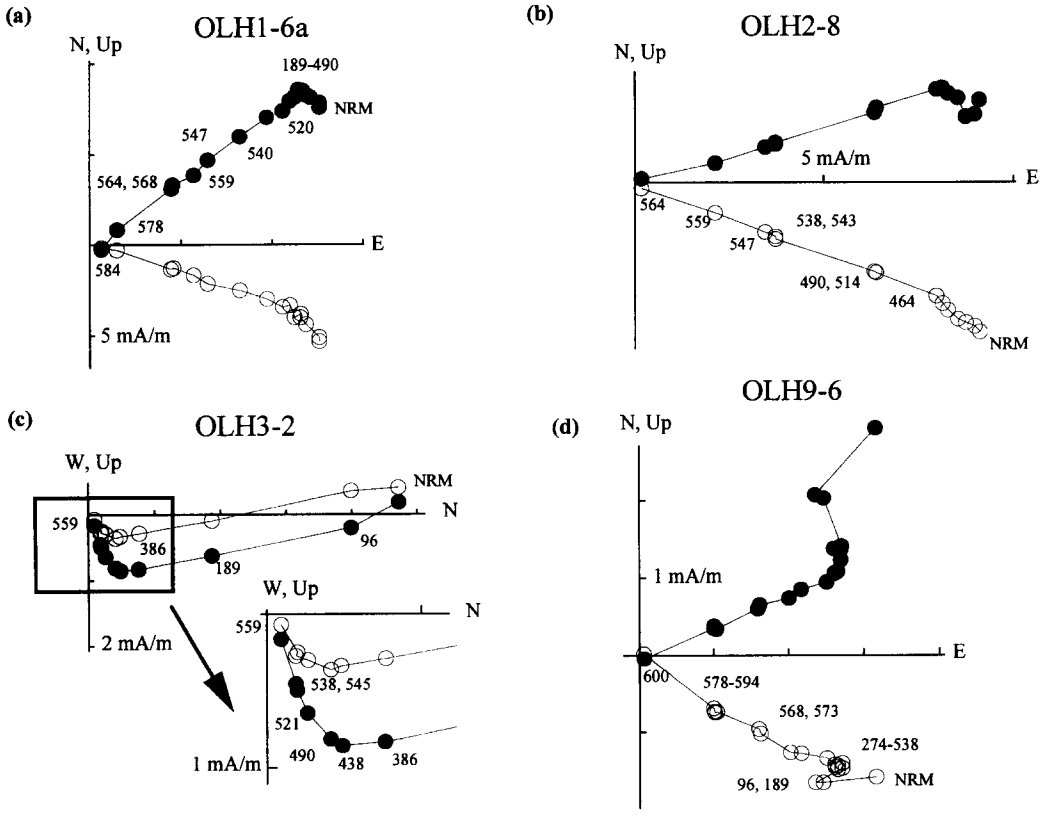


Fig. 5. Representative tilt-corrected orthogonal vector endpoint diagrams of thermal demagnetizations that display the characteristic direction of the Lawrence Head Volcanics. Closed (open) symbols represent projections onto the horizontal (vertical) plane. Numbers refer to temperatures of demagnetization steps in °C.

magnetizations. The characteristic directions are dominantly northeasterly and shallow downward (Fig. 5).

The representative diagram for site OLH3 (Fig. 5c) shows similar directional behavior, except that the viscous overprint represents a greater proportion of the total NRM and that there is a slight curvature in the transition between the two vectors. The assertion that the first component is viscous is supported by the agreement of its in-situ direction ($D/I = 358°/69°$) with that expected for the present geomagnetic field ($D/I = 343°/72°$). Samples from OLH9 (Fig. 5d) have an initially random magnetization that is removed by the first demagnetization step ($\sim$ 100°C) to reveal the two components common to other Lawrence Head sites. The higher temperature component is defined by eight demagnetization steps between 560°C and 600°C.

Site-mean directions calculated from stable-endpoint directions (Table 1) are shown in Fig. 6. In-situ site means scatter, but begin to cluster better upon correction for tilt, and cluster fairly well after a further correction is made by rotating site means so that strike trends are aligned. We will return below to the tilt- and strike-corrected results.

Fig. 7 displays demagnetization diagrams and stereonets with great-circle trajectories as examples of the data used to compute great-circle poles for seven sites (Table 2). No samples from site 7 (Fig. 7c, d) reach stable endpoints, but they show clear great-circle trends; poles to site-mean great circles for site 7 and the other six sites are plotted in Fig. 8. In in-situ coordinates, these poles lie themselves on a great circle, showing that the magnetizations have a common component in in-situ coordinates that is close to the local present-day geomagnetic field di-

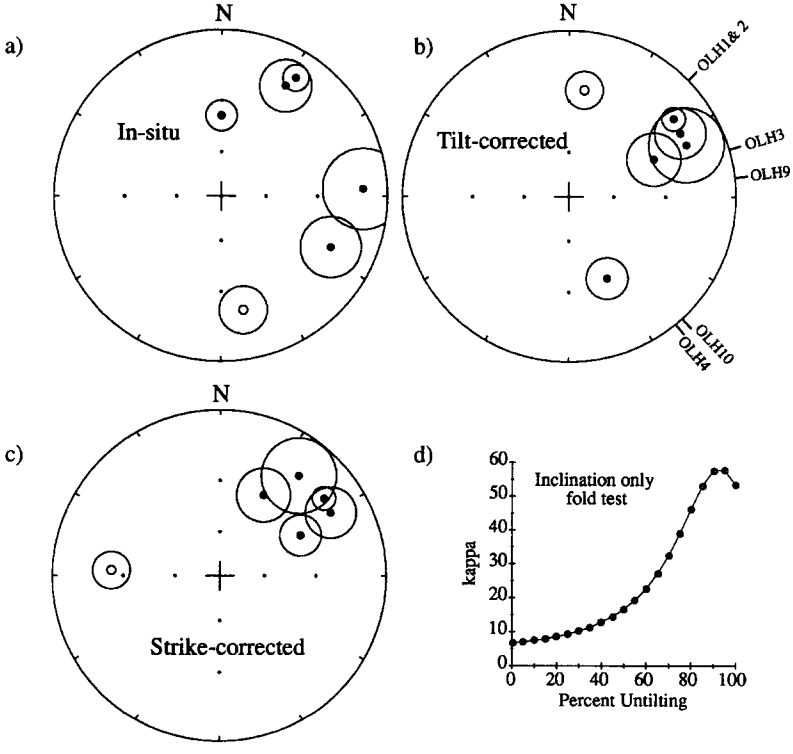


Fig. 6. Site mean directions and associated cones of confidence (α_{95}) for the Lawrence Head Volcanics plotted on equal-angle stereonets in (a) in-situ, (b) tilt-corrected and (c) tilt- and strike-corrected coordinates [reference strike for (c) is 045°]. The inclination-only test (d) peaks at 90% tilt correction, but there is no significant difference with 100% correction (i.e., a pretilting magnetization.

rection. Upon tilt-correction, the site-mean poles form a broad grouping in the northwesterly quadrant, whereas upon strike correction (to be discussed below) the poles lie along a great circle that defines an overall formation-mean pole with an easterly and shallow down direction.

5.3. Fold and strike tests

It is well known that simple tilt corrections, performed by rotating about the strike, yield directions with correct inclinations but with possibly deviating declinations, for instance when folds are plunging.

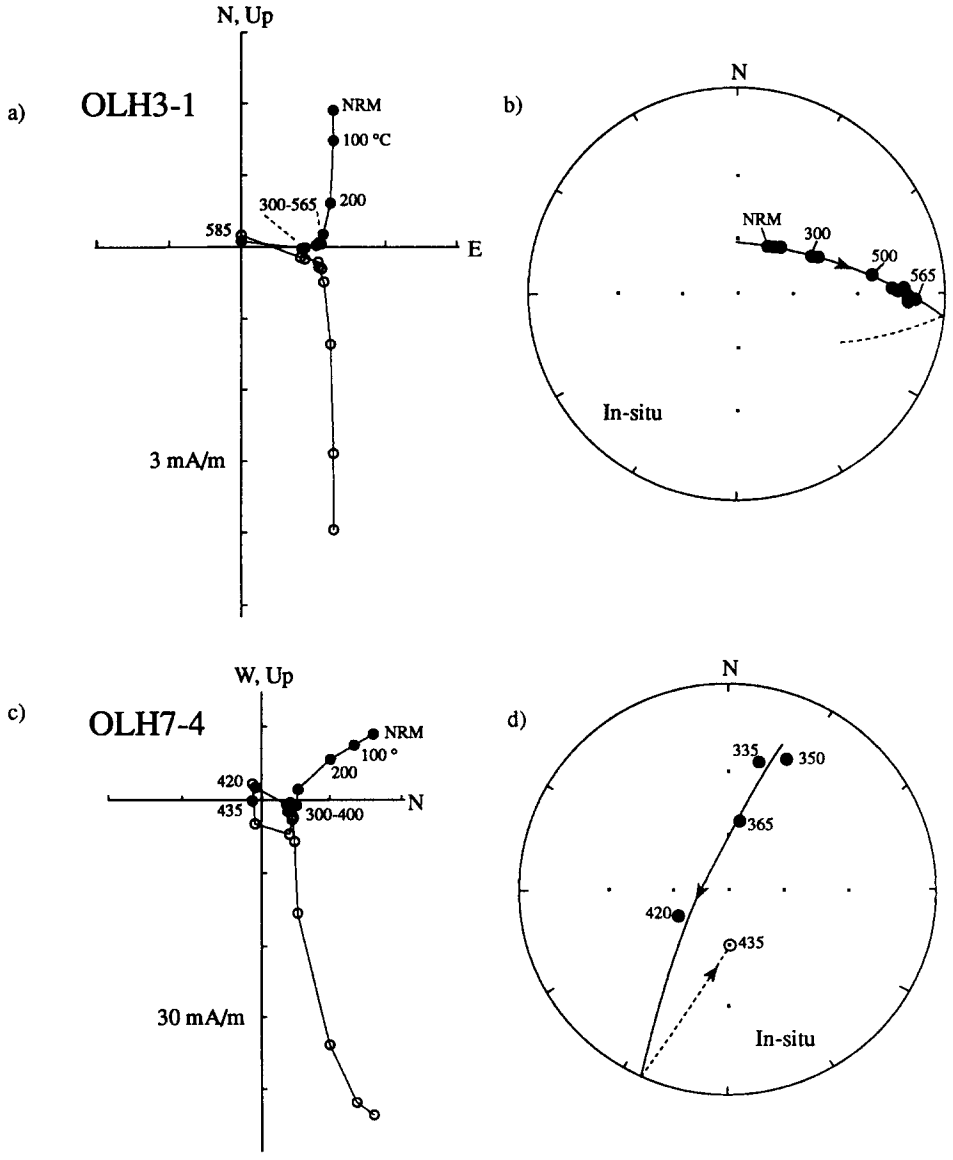


Fig. 7. Typical thermal demagnetization diagrams as in Fig. 5. Numbers refer to temperatures of demagnetization steps in °C.

Table 2
Poles to remagnetization circles from the Ordovician Lawrence Head Volcanics, Central Newfoundland

Site	Bedding S/D	*n/N*	In-situ T/P	Tilt- corrected T/P	Strike- corrected T/P	*k*	α_{95}
OLH1	045/80	5/12	129/19	303/61	303/61	81.4	8.5
OLH2	045/88	5/11	131/26	308/62	308/62	66.5	9.4
OLH3	253/80	9/10	181/22	289/69	261/69	44.5	7.8
OLH4	320/80	6/23	265/07	318/55	223/57	13.1	19.6
OLH7	320/87	10/31	094/23	286/40	191/40	61.1	6.2
OLH9	263/84	11/13	209/06	260/62	222/62	16.7	24.5
OLH10	317/99	11/15	076/05	323/61	231/61	35.5	7.8
Pole			331/64		082/22		

Notes: Symbols are the same as in Table 1. T and P are the trend and plunge of the pole to the remagnetization circle.

Moreover, if sites have rotated before folding occurred, as is common in some curved orogenic belts (e.g., Eldredge et al., 1985), then tilt-corrected site means show girdle distributions with similar inclinations. A strike test was devised (Perroud, 1983) in which a magnetization can be determined to be pre-rotational if the declinations agree better when the curved belt is straightened or, in other words, when all strikes are aligned. In the case of the neighboring Moreton's Harbour Group (Johnson et al., 1991), the magnetization appears to be pre-folding as well as pre-rotational, therefore we tested

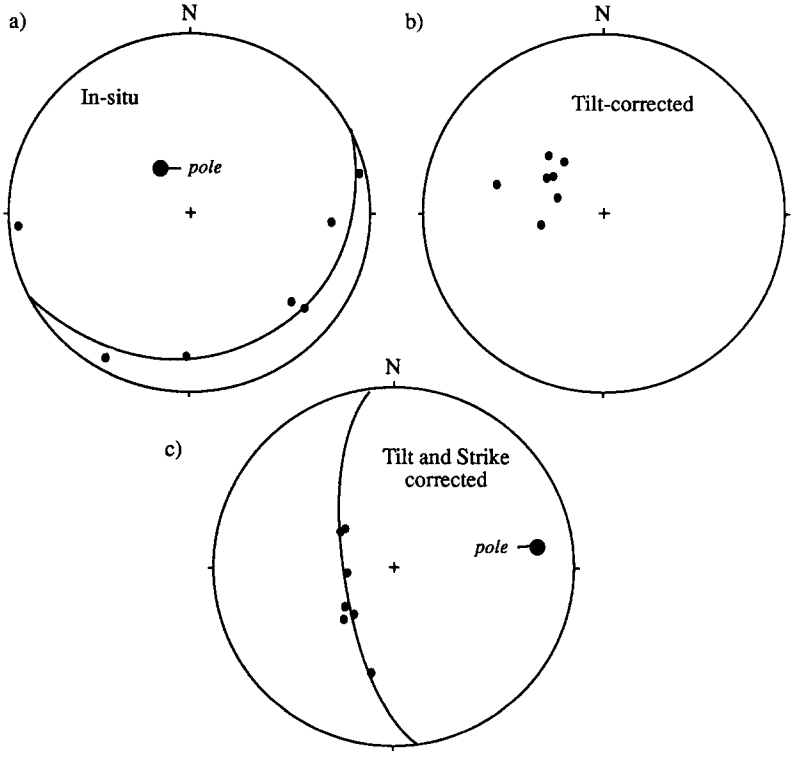


Fig. 8. Poles to great-circles (small dots) for seven sites listed in Table 2. The tilt- and strike corrected orientation in (c), 082°/22°, gives a direction that is in close agreement with stable endpoint analysis (Fig. 6c; Table 1).

whether the same might be true for the Lawrence Head Volcanics. The procedures followed were: (1) to determine tilt-corrected directions; (2) test whether the magnetization is pre-folding by performing an inclination-only fold test (Pares et al., 1994); (3) rotate the site-mean directions by aligning the strikes, using an arbitrary reference strike of 45°E; and (4) test whether the site-means cluster significantly better after such rotations.

The inclination-only fold test is positive (Fig. 6d), although it peaks at 90% untilting; however, this is not distinctly different from 100% untilting. Given the age of the folding, the magnetization must have been acquired before mid- to late Silurian times. The strike-corrected site means cluster better than the directions corrected for tilt only (cf. Fig. 6c, b) with k improving from 2.7 to 14.1 (Table 1). Dual-polarity directions are observed, although these are not quite antipodal, probably due to uncertainties in the strike correction. The strike test illustrates that the magnetization predates rotations about vertical axes that likely occurred during accretion of the terrane and strike-slip movements in the area. Similar rotations were observed in neighboring areas (Johnson et al., 1991; Van der Voo et al., 1991).

It is noteworthy that the poles to site-mean great circles show a much better definition after strike correction (Fig. 8c) than before (Fig. 8b), and only then define an overall formation mean pole that is easterly and shallow down. This direction is roughly similar to the end-point directions of Fig. 6c. The great-circle part of the directional analysis (Table 2) uses for a large part the same samples as those used for the stable-endpoint directions, and thus can only be used as supportive evidence that an overall east-northeasterly shallow down direction characterizes the sites with NE strikes in the Lawrence Head volcanics. As listed in Table 1, the stable endpoint

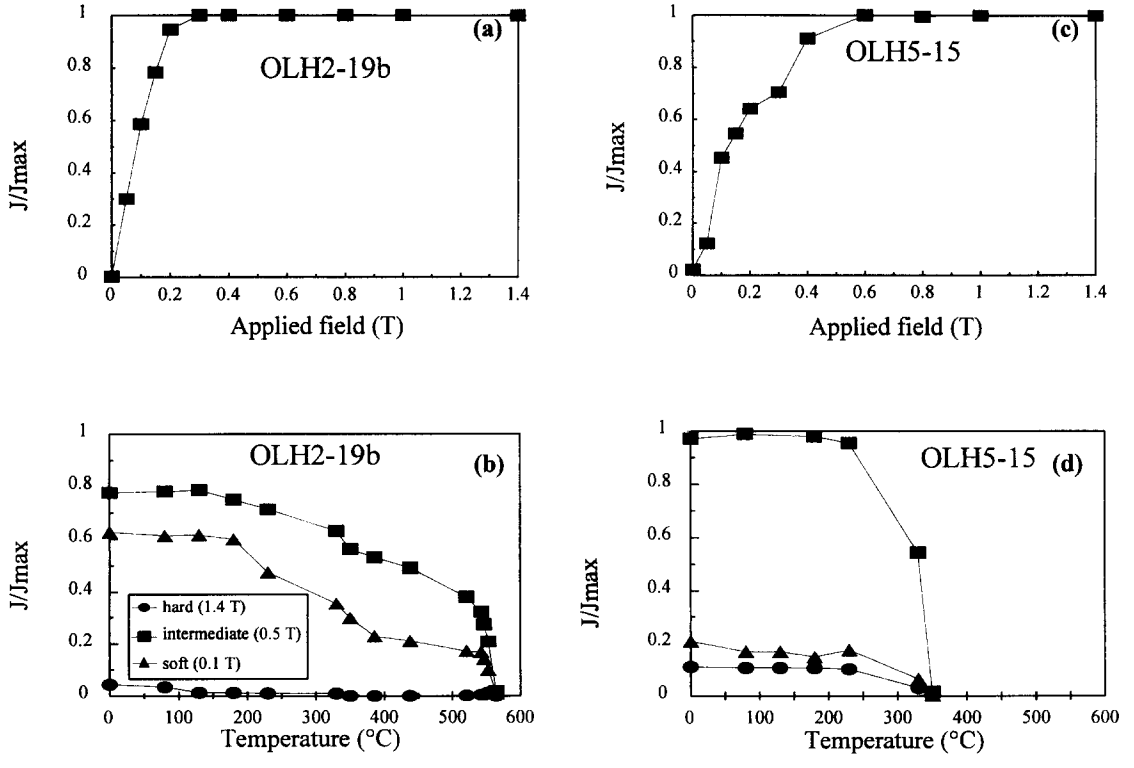


Fig. 9. IRM acquisition (a, c) and thermal demagnetization of multi-component IRM (b, d) for representative volcanic (OLH2-19b) and sedimentary (OLH5-15) samples. Except for (b), magnetization is normalized to the maximum resultant magnetization. In basalt pillows it is likely carried by single or pseudo-single domain magnetite, but in the sediments of site 5 it likely resides in pyrrhotite.

directions define a formation mean direction of $D/I = 56/+23$ ($k = 14.1$, $\alpha_{95} = 18.5$, $N = 6$), with a corresponding paleo-(south)pole at 30.9°N, 54.9°E ($d_p = 10.4$, $d_m = 19.7$).

5.4. Rock magnetic analyses

One AF-treated sample from each site was used for rock magnetic experiments to characterize the magnetic carriers and how they may differ between sites. The first procedure involved stepwise acquisition of an IRM. Cores were then given an IRM of 1.4 T, 0.5 T and 0.1 T along three mutually perpendicular axes before stepwise thermal demagnetization. By matching remanent coercivity with unblocking temperature for any of the three orthogonal components, the magnetic mineralogy can be more precisely characterized.

Fig. 9 displays the results of these procedures for two samples, one representing basalt pillows (OLH2-19b), the other (OLH5-15) representing the sedimentary beds at site 5. The IRM acquisition in the basalt pillows (Fig. 9a) is characteristic of magnetite, which saturates in a field of 0.3 T. This assertion is further confirmed by the thermal demagnetization of the multi-component IRM. The magnetization is dominated by the intermediate component and unblocking is complete at approximately 580°C (Fig. 9b). The hard shoulder indicates that much of the magnetite likely has a low titanium content. The soft component is probably multi-domain magnetite and is likely responsible for the present-day-field overprint displayed in orthogonal vector diagrams.

The sediments of site OLH5 display magnetic behavior different from that of the volcanic sites, and characteristic of pyrrhotite. The IRM acquisition is complete at 0.6 T (Fig. 9c), although the small gradient that can occur in pyrrhotite above this level may still be unresolved in a field up to only 1.4 T. Thermal demagnetization of the multi-component IRM yields results that could be interpreted as titanomagnetite or titanomaghemite, were it not for the high coercivities indicated by the IRM acquisition. Taken together, these experiments indicate that the magnetic carrier in sediments is pyrrhotite, which reveals only the present-day Earth's field direction.

6. Discussion

The age of the magnetization is bracketed by the mid-Arenigian to Llanvirnian age of the rocks (Hibbard et al., 1977; Williams et al., 1992) and the mid- to late Silurian age of deformation (Helwig, 1967; Blewett and Pickering, 1988; Lafrance and Williams, 1992; O'Brien, 1993). During this time interval the margin of North America to which these rocks were accreted was located in the Southern Hemisphere and facing south, so it is logical to assume that the characteristic direction of the Lawrence Head Volcanics represents the geomagnetic field in (low) southerly latitudes, thus implying that it was acquired in a reversed polarity field. The paleolatitudes of western Newfoundland (i.e., the Laurentian margin of Iapetus) were low to equatorial during this interval (Fig. 10). We therefore conclude that the Lawrence Head Volcanics acquired its remanence at 12° ± 10°S, adjacent to the Laurentian margin of the Iapetus Ocean, and that the remanence is primary because of the positive inclination-only test, and because (Silurian, Early Devonian) paleogeographic positions of western Newfoundland are at higher southerly paleolatitudes. This tectonic setting contrasts with the conclusion that the Lawrence Head Volcanics formed as a result of back-arc rifting at the *southern* margin of Iapetus (Hughes and O'Brien, 1994), which they based on the vergence of thrusting

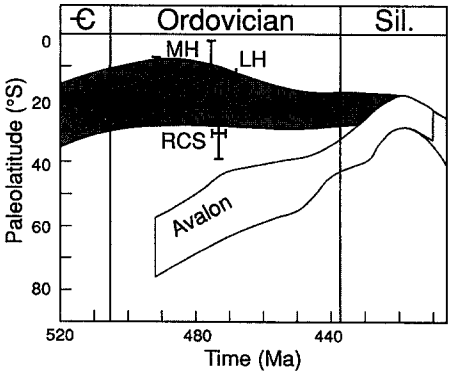


Fig. 10. Paleolatitudes of the Lawrence Head (*LH*), Moreton's Harbour (*MH*) and Robert's Arm–Chanceport–Summerford (*RCS*) studies plotted as a function of time. Included are paliolatitudes calculated for a reference site at 305°E, 49°N from the North American reference path (Van der Voo, 1990) and five poles for Avalonia obtained in southern Britain (Table 3).

in the area. The observed low paleolatitude from our study differs significantly from that of Avalon/Gondwana, which was 50°S or higher in the Middle Ordovician (Fig. 10; Van der Voo, 1993). Finally, the ca. 12°S paleolatitude reported for the Lawrence Head Volcanics is similar to that of 11°S ± 4° reported by Johnson et al. (1991) for the Moreton's Harbour Group, suggesting that they were part of the same near-Laurentian arc–back-arc complex.

6.1. Tectonic implications

The results of this study in combination with those of the Moreton's Harbour and Robert's Arm–Chanceport–Summerford Groups (Fig. 10 and Table 3) show that the closure of Iapetus at the Laurentian margin caused a complex aggregation of terranes that came from different paleolatitudes. The out-of-sequence paleolatitudes that these three studies display on the Fortune Harbour Peninsula pose a geometric problem to which we propose two solutions (Fig. 11). In the first scenario the RCS terrane rafts in from 31°S and is thrust over the accreted Lawrence Head–Moreton's Harbour arc complex along a fault that subsequently becomes rotated into a tight synform. While invoking a single klippe scenario (Fig. 11a) is perhaps the simplest geometric solution, a variation involving out-of-sequence thrusting that is observed by recent field mapping in the area (e.g., O'Brien, 1990) is also permissable.

In the second scenario, the RCS terrane first is accreted to the Moreton's Harbour–Lawrence Head Arc, but later strike-slip motion along the Luke's Arm Fault causes the Lawrence Head Volcanics and Exploits Group to be juxtaposed against the southern exposure of the RCS Terrane (Fig. 11b). Structural

Table 3
Paleolatitudes of Ordovician Newfoundland central mobile belt studies and bordering continents

Formation or reference pole	Age (Ma)	South pole (Europe)	South pole (N. Am)	Pl (°S)	d_p (°)	Reference
North America						
Eu, Em, E	506–542		9S/338E	25	10	Van der Voo, 1990
Ol/m, Ol	468–505		15S/344E	17	11	Van der Voo, 1990
Ou/Sl, Ou, Om	430–467		16S/324E	23	6	Van der Voo, 1990
Su/Sm	415–429		19S/403E	22	4	Van der Voo, 1990
Su/Dl	398–414		3S/277E	33	7	Van der Voo, 1990
Newfoundland						
CMB						
Moreton's Harbour	505–478		29S/315E	11	4	Johnson et al., 1991
Lawrence Head	483–468		31N/55E	12	10	This study
RCS	475–471		8S/317E	31	8	Van der Voo et al., 1991
Avalon						
Treffgarne Volc.	505–478	56N/306E	57N/268E	67	9	Trench et al., 1992
N. Builth Sed./volc./intrusives	480–460	18N/13E	19N/335E	52	10	McCabe et al., 1992
Borrowdale Volc.	460–440	8N/6E	9N/328E	46	8	Channell and McCabe, 1992
Dunn Point Volc.	419–449		2S/310	39	4	Johnson and Van der Voo, 1990
E. Mendip Inlier	420	13N/271E	14N/233E	22	5	Torsvik and Trench, 1992
Old Red Sandstone	410	7S/307E	6S/269E	27	5	Channell et al., 1992

Poles for Avalon reported from studies conducted in Southern Britain are rotated 38° clockwise from European (Europe) to North Americal (N Am) coordinates about a Euler pole located at 88.5°N, 27.7°E (Bullard, 1965, Van der Voo, 1990) to account for the closure of the Atlantic Ocean. Poles for Laurentia include Greenland and northern Scotland (Van der Voo, 1990). Pl refers to paleolatitude calculated for a point at 49°N, 305°E and is reported in degrees south. d_p refers to 95% confidence interval in paleolatitude.

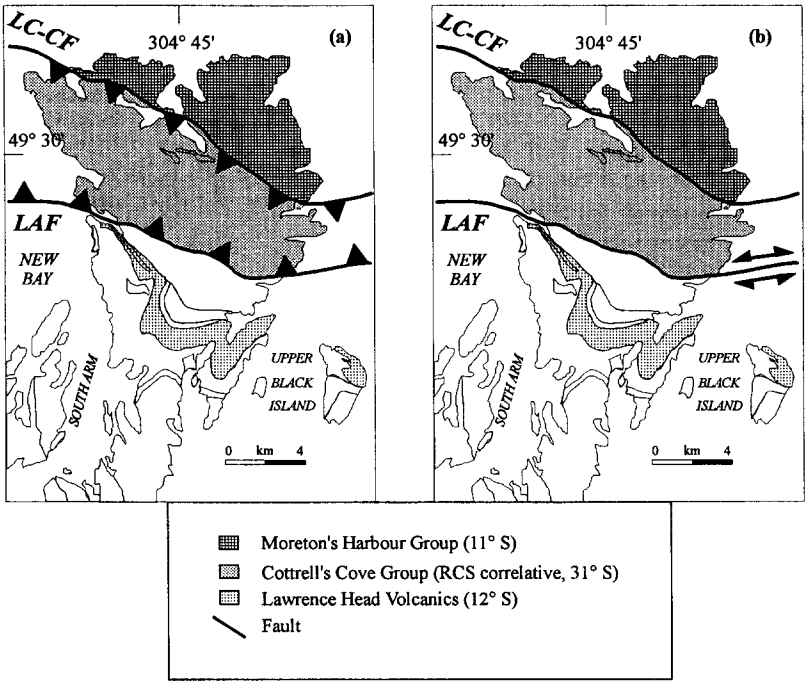


Fig. 11. Generalized terrane map of the Fortune Harbour Peninsula (as in Fig. 2) showing two possible interpretations for the present-day geometry and juxtaposition of paleolatitudinally distinct terranes. In (a) the RCS Terrane is interpreted as a (folded) klippe overriding the Lawrence Head Volcanics and Moreton's Harbour Group, whereas in (b) strike-slip motion (double-arrows) modifies the sequence after the RCS Terrane accreted.

studies in the area have concluded opposite senses of shear along faults. Blewett and Pickering (1988) emphasize sinistral shear, whereas Lafrance (1989) concludes dextral motion, and each argues that the opposite shear sense is only observed where a later cleavage overprint is mistakenly used. Either shear sense is possible based upon the local orientation of two temporally discrete kinematic axes inferred by O'Brien (1993). These are first northeast–southwest and later northwest–southeast, respectively, consistent with both sinistral and dextral shear. We cannot argue vigorously for any one local shear sense based on the paleomagnetic data. The Summerford samples from sites located immediately to the south and east of the Luke's Arm Fault on New World Island display high paleolatitudes and require that in the vicinity of New World Island to the east, major shearing is located on a fault to the south of the Summerford Group, perhaps along the south shore of New World Island where a Silurian accretionary complex is faulted against the Dunnage Melange (van der Pluijm, 1986).

A two-step scenario of thrusting followed by wrench motions was suggested by Arnott et al. (1985) to account for sedimentation patterns throughout the Notre Dame Bay area, and in structural studies of Lafrance and Williams (1992), O'Brien (1993) and others, who find that thrusting precedes wrench motions in the region. Our previous studies limit thrusting to early Silurian times or older (van der Pluijm et al., 1993), whereas the current study constrains the timing of strike-slip motion as occurring after docking of the RCS Terrane, i.e., post-Llandovery. Regardless of the details of the thrusting and strike-slip scenarios or their combination, these interpretations explain the apparently conflicting locations of the Red Indian Line according to Williams et al. (1988) and van der Pluijm et al. (1990) by invoking structural repetition of this major tectonic boundary in the northcentral Newfoundland Appalachians.

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